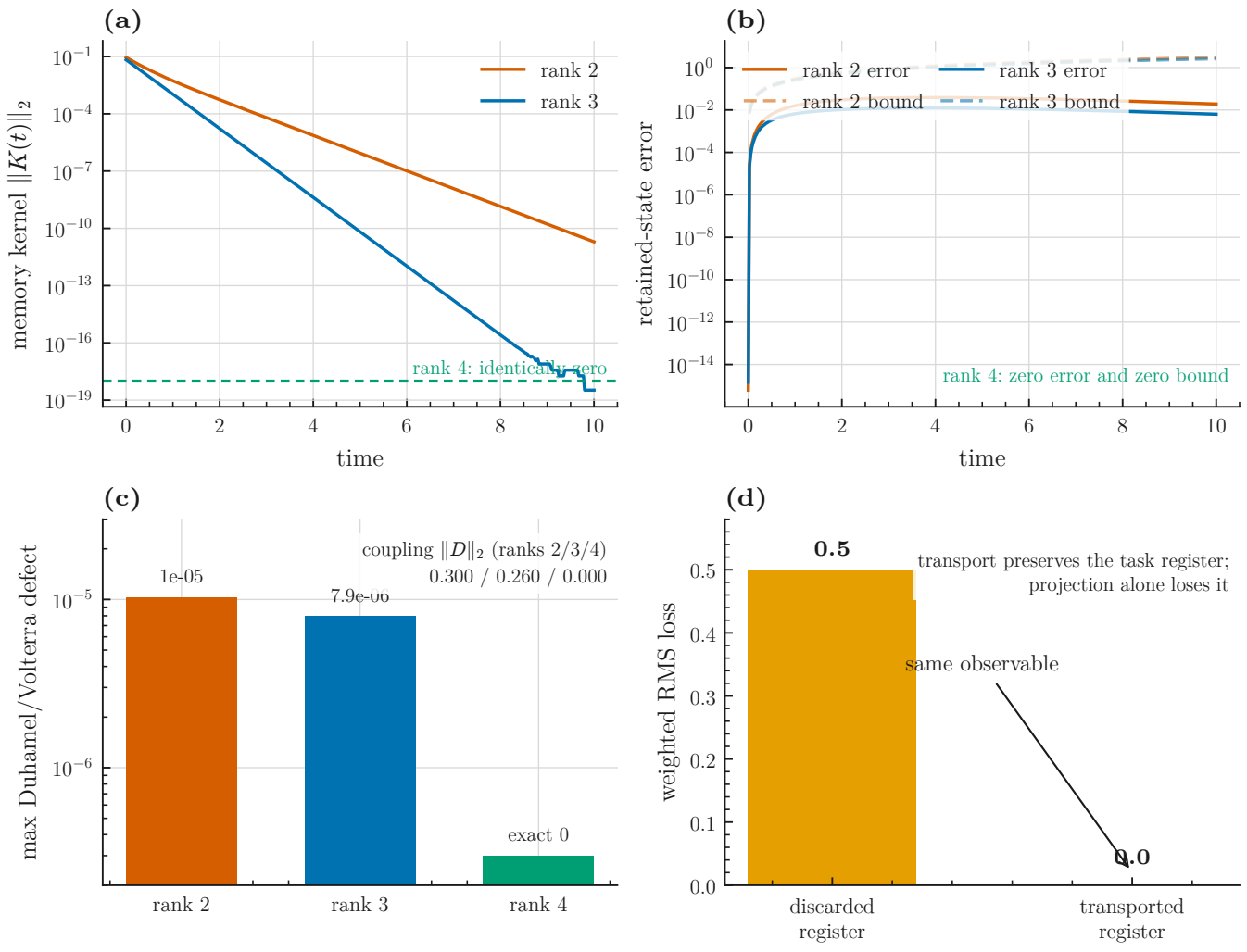
(a)
memory kernel $\|K(t)\|_2$
rank 2
rank 3
rank 4: identically zero
time
(b)
retained-state error
rank 2 error
rank 3 error
rank 2 bound
rank 3 bound
rank 4: zero error and zero bound
time
(c)
max Duhamel/Volterra defect
coupling $\|D\|_2$ (ranks 2/3/4)
0.300 / 0.260 / 0.000
1e-05
7.9e-06
exact 0
rank 2
rank 3
rank 4
(d)
weighted RMS loss
0.5
transport preserves the task register;
projection alone loses it
same observable
0.0
discarded register
transported register